

FEHMARN, Germany

WMO: 100550

Lat: 54.5283N

Lon: 11.0606E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-4.2	-9.2	1.7	-4.1	-7.5	2.0	-3.4	18.3	6.7	16.5	5.9	4.3	120	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	24.7	19.1	23.1	18.7	21.8	18.1	20.2	23.2	19.3	22.0	18.6	21.0	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.8	21.7	18.3	13.2	20.8	17.5	12.5	20.1	57.9	23.1	55.0	22.0	52.6	21.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.4	13.6	12.1	-7.4	28.6	2.0	2.0	-8.8	30.1	-10.0	31.2	-11.1	32.3	-12.6	33.7		
			-8.2	21.7	1.8	0.8	-9.5	22.3	-10.5	22.8	-11.6	23.3	-12.9	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	1.7	1.8	3.7	7.5	11.9	15.2	17.8	18.0	15.0	10.5	6.2	2.8
	DBStd	6.54	3.46	3.15	2.97	2.63	2.37	2.21	2.25	2.09	2.29	2.76	2.93	3.36
	HDD10.0	1139	256	229	196	83	9	0	0	0	0	27	116	223
	HDD18.3	3311	514	462	454	324	201	98	37	32	102	242	363	482
	CDD10.0	919	0	0	0	9	67	155	243	248	151	44	3	0
	CDD18.3	49	0	0	0	0	0	3	22	22	3	0	0	0
	CDH23.3	123	0	0	0	0	4	14	53	44	7	0	0	0
	CDH26.7	13	0	0	0	0	1	2	5	4	1	0	0	0
Wind	WSAvg	5.7	5.9	6.4	6.1	5.3	5.1	5.5	5.3	5.2	6.1	6.1	5.8	6.0
Precipitation	PrecAvg	580	46	36	33	31	48	57	59	63	49	52	54	50
	PrecMax	730	90	76	66	71	92	121	135	190	156	104	127	96
	PrecMin	381	1	1	11	2	13	25	17	16	17	29	11	20
	PrecStd	90	21	18	15	15	22	22	32	44	30	18	31	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	10.1	12.9	18.0	22.1	25.4	27.0	26.7	24.1	18.1	13.2	10.8
		MCWB	8.5	8.2	9.3	13.6	16.8	19.3	20.3	19.7	17.2	15.4	11.7	9.5
	2%	DB	8.2	8.1	10.2	14.9	18.6	22.2	24.7	24.4	21.3	16.3	12.1	9.0
		MCWB	7.1	7.0	8.2	11.6	14.3	17.8	19.3	19.3	17.1	14.2	10.8	7.9
	5%	DB	7.2	7.0	8.7	13.1	17.1	20.2	23.2	22.8	19.7	15.1	11.1	7.9
		MCWB	6.2	5.9	7.3	10.4	13.7	16.7	18.9	18.8	16.4	13.4	9.8	7.1
	10%	DB	6.2	6.0	7.5	11.5	15.7	18.7	21.8	21.5	18.6	14.2	10.1	7.1
		MCWB	5.4	5.3	6.2	9.5	12.9	15.7	18.2	18.3	15.8	12.7	8.9	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.4	10.3	13.9	17.1	19.9	21.7	21.2	18.7	16.0	12.2	10.0
		MCDB	9.6	9.4	11.7	17.3	20.8	23.9	25.3	24.5	21.4	17.5	12.9	10.3
	2%	WB	7.3	7.2	8.6	12.0	15.4	18.3	20.4	20.1	17.7	14.6	11.0	8.1
		MCDB	8.0	8.1	9.9	14.0	17.7	21.2	23.4	23.1	20.4	15.8	11.9	8.7
	5%	WB	6.3	6.1	7.4	10.7	14.3	17.2	19.4	19.3	16.9	13.7	10.0	7.2
		MCDB	7.0	6.7	8.6	12.5	16.3	19.4	22.1	21.8	19.4	14.8	10.9	7.8
	10%	WB	5.5	5.3	6.3	9.7	13.3	16.2	18.6	18.6	16.0	12.9	9.1	6.3
		MCDB	6.2	5.9	7.3	11.2	15.2	18.3	21.3	20.9	18.3	14.1	9.9	7.0
Mean Daily Temperature Range	5% DB	MDBR	2.9	2.9	3.9	5.4	5.7	5.4	5.4	5.1	4.5	3.8	2.9	2.9
		MCDBR	3.4	3.6	6.3	8.5	8.4	8.5	8.3	7.3	7.1	4.9	3.4	3.1
		MCWBR	3.2	3.2	4.7	5.6	5.3	5.1	4.3	3.9	4.0	3.6	3.0	3.0
	5% WB	MCDBR	3.2	3.5	6.0	7.6	7.6	7.7	7.3	6.3	6.4	4.5	3.3	3.1
		MCWBR	3.1	3.2	4.7	5.3	5.3	5.1	4.2	3.8	4.0	3.5	3.1	3.1
Clear-Sky Solar Irradiance	taub	0.320	0.341	0.371	0.407	0.398	0.395	0.414	0.413	0.385	0.361	0.340	0.310	
	taud	2.300	2.317	2.264	2.214	2.339	2.378	2.338	2.351	2.410	2.424	2.361	2.287	
	Ebn at Noon	615	736	795	815	847	854	827	803	779	709	589	541	
	Edh at Noon	58	79	104	124	116	113	116	108	90	71	55	48	
All-Sky Solar Radiation	RadAvg	0.48	1.10	2.41	4.14	5.28	5.60	5.22	4.20	3.04	1.58	0.64	0.35	
	RadStd	0.07	0.18	0.23	0.50	0.58	0.37	0.59	0.29	0.30	0.20	0.08	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	N/A

Nomenclature: See separate page